

Project 1: Predicting mortality risk after elective general surgery

Project name	Predicting mortality risk after elective general surgery
Project owner/sponsor	Nicola Rennie
Project description	Sleep deprivation can cause cognitive impairment and performance decline, and there is no reason to believe that hospital staff are immune to such effects. Like many professionals, hospital staff are likely to become more fatigued during the working day, as the week goes on, and at times of higher workload. These time-related factors may lead to an increase in fatigue-related decision errors which could worsen patient outcomes. The aim of the project is to better understand the risks associated with general elective surgery by developing a model to predict 30-day mortality, after accounting for patient characteristics and co-morbidities in addition to the effects of surgery timing. This project is open to many avenues of research, though two potential research questions may be: (i) After accounting for the baseline risk of the surgery, do operations that occur towards the end of the working week result in higher risks of (a) 30-day mortality; and (b) in-hospital complications? (ii) How does this change in risk vary across procedure category? This data is well suited to the application of statistical models, but machine learning techniques (or other appropriate methods of the students' choice) would also be suitable. Students are also encouraged to consider the limitations of the data and consider which additional information could be relevant to collect.
Project Data	The data and data dictionary can be freely accessed at this link: https://www.causeweb.org/tshs/surgery-timing/

Organization details

Organization name	Centre for Health Informatics, Computing and Statistics
Main contact name	Nicola Rennie
Main contact position	Lecturer in Health Data Science

Project 2: Predicting a patient's length of stay

Project name	Predicting a patient's length of stay
Project owner / sponsor	NHS Wrightington, Wigan and Leigh
Project description	With varying demand but a fixed capacity of beds, bed managers at WWL have the difficult job of making critical decisions in facilitating the effective use of acute hospital beds, often relying on their own experience, initiative, supplementary data, and forecasts during this process. In recent times, tools taking advantage of artificial intelligence and other statistical methods have been developed with the objective of assisting bed managers in making these decisions. One such method is predicting an individual patient's length of stay – the period from admittance to a ward, to discharge from the hospital. Predicting a patient's length of stay can conceivably be completed by considering a range of patient information, including their demographics, diagnoses, and comorbidities. Task: • Explore the dataset and summarise it. • Build and evaluate a model which predicts patient length of stay in days or hours. • Suggest additional data which could be used to improve your model. • Using these results as context, describe the importance of selecting an appropriate date range for building a model.
Project Data	WWL_NHS.csv attached contains a subset of the electronic patient record (EPR) data that are collected for each patient admitted into the hospital. Definitions of ambiguous variables in the data set are shown below: - <i>Spell_episode_los</i> = the length of stay of an individual patient's "spell" (A Hospital Provider Spell is the total continuous stay of a patient using a Hospital Bed) - <i>Admission</i> = entry time point into a hospital bed. Beginning of spell - <i>Discharge</i> = exit time point out of hospital bed. End of spell. - <i>Readmission_flag_28_days</i> = Binary variable. Has a patient had a previous spell within 28 days. 1 = yes; 0 = no. - <i>Dementia_diagnosis_flag</i> = Binary variable. Patient has dementia. 1 = yes; 0 = no.

	- <i>Spell_primary_diagnosis</i> = primary diagnosis is the International Classification of Diseases (ICD) code used to identify the primary diagnosis. - <i>Spell_secondary_diagnosis</i> = secondary diagnosis is the International Classification of Diseases (ICD) code used to identify the secondary diagnosis. - <i>Spell_dominant_proc</i> = the primary procedure carried out to the patient. - <i>doctor_foster_comorbidity_score</i> = scoring system for assessing patient frailty.
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Organisation Details

Organisation name	NHS Wrightington, Wigan and Leigh
Main contact name	Dr Thomas Ingram
Main contact position	Head of Data Science